Inhalerbuddy

Executive Summary: Inhalerbuddy LLC

Overview

Inhalerbuddy is a connected health startup that is passionate about changing the way that families, schools and pharmacies think about childhood asthma. We are a team of three students at our university and we have formed an LLC and built a working prototype. This executive summary will describe the problem, our solution, the market, our business model, the competition, our team, our milestones and what we are asking for.

The Problem

Asthma is a very common condition among children and it is a problem that effects millions of families. Many kids with asthma do not take there medicine the way that the doctor said, and parents often do not know when a inhaler was used or not used. Sometimes a child will use the rescue inhaler a lot more then normal in the days before a attack, but nobody notices because there is no good way to see the pattern. This can lead to emergency room visits, missed school days, and stress for the whole family. We beleive that this is a big problem that is not being solved well today and that technology can help. Parents are busy and they want peace of mind about there child's health.

The Solution

Our solution is Inhalerbuddy, which is a end-to-end platform that leverages Internet of Things sensing, mobile software and human pharmacist support to create a seamless asthma management experience. It has three pieces.

First, the Inhalerbuddy cap. This is a small electronic cap that snaps on to the top of most common inhalers. It has a sensor that detects when the inhaler is pressed and it sends the time to the phone using wireless. The cap is lightweight and designed to be kid friendly with fun colors.

Second, the Inhalerbuddy app. The app shows caregivers a dashboard of when the medicine was used. If the rescue inhaler is used more often then usual, the app sends a alert to the caregivers so they can check on the child, call the doctor or take other steps before symptoms get worse. The app also has reminders for daily controller medicine.

Third, the pharmacist check-in service. Once in a while a pharmacist reviews the data and checks in with the family by message or call to answer questions and make sure the child is using the inhaler properly. We think this human touch is what makes us different and gives parents trust.

Together these three pieces create a holistic solution that is proactive instead of reactive.

Market

The market for our product is very large. There are millions of children with asthma in the United States alone and the market for digital health is growing very fast. If we capture even a small percent of families with asthmatic children, this would be a very large business. We also believe there are opportunities internationally and with other conditions besides asthma, such as other respiratory conditions and chronic disease management. Our target customers are parents of children with asthma, and secondarily schools, pediatricians and pharmacies. We have not yet done a formal market study but we have talked to a few parents and friends who thought the idea was good.

Business Model

We plan to make money in several ways, and we are still figuring out which will work best. One

option is to sell the cap as a one time purchase to parents at a price of around thirty to fifty dollars. Another option is a monthly subscription for the app and the pharmacist service, maybe five to ten dollars a month. We could also partner with pharmacies, insurance companies or health systems who might pay us to offer Inhalerbuddy to their patients or members. There is also the possibility of selling anonymized data to researchers or pharmaceutical companies, which could be very valuable. We think a mix of these revenue streams will make the business sustainable. Pricing will be determined after we do more testing with customers.

Competition

There are some other companies that make smart inhaler sensors and asthma apps. However we feel that our product is better because it combines hardware, software and a pharmacist service in one package and because it is focused on kids and caregivers. Other products are mostly for adults or for the patient only. Also, our design is more fun and friendly for children. We believe that no competitor offers the full solution that we do, and we have a first mover advantage in the pharmacist check-in piece. We will also build a strong brand which will be hard to copy.

Team

Brandon Tilley is our CEO and he is a junior studying business. He is a natural leader and has always been interested in health and entrepreneurship. Marisol Quenneville is our CTO and she is a senior studying electrical engineering, and she designed and built the prototype cap. Devraj Ostrowski is our head of product and he is a junior studying computer science, and he built the first version of the app. Our team is hardworking, passionate and committed to making Inhalerbuddy a success. We do not yet have a pharmacist or a clinician on the team but we plan to find advisors, and one of our relatives is a nurse who has given us some feedback.

Milestones and Financials

So far we have reached two main milestones. We defined our core business concept and we built a prototype of the cap and a early version of the app. We have also formed a LLC. The prototype works in the lab and can detect when it is pressed and send the data to the phone, although the battery life and the fit on different inhalers still needs work. We have not yet tested with real patients or children and we have no customers or revenue at this time. We have also not yet made a website. Our expenses so far have been small and paid for by the founders out of pocket, mostly for electronic parts and a few app tools. Over the next year, we expect to need money for manufacturing, regulatory work, app development, pharmacist staffing and marketing. We think we could reach substantial revenue in about three years if things go well, though we have not yet built a detailed financial model.

What We Are Asking For

We are applying to this competition for several reasons. First, we hope to win prize money which we would use to improve the prototype and begin a small pilot. Second, we are looking for mentorship from experienced entrepreneurs and healthcare professionals who can help us with our business model and with navigating regulations. Third, we would like introductions to potential partners such as pharmacies, pediatric clinics and schools. Finally we want feedback from the judges on how to make our idea stronger. We are excited about the opportunity and we thank you for considering our application. With your support we beleive Inhalerbuddy can make a real difference in the lives of children with asthma and there families.